

# To Intervene or not to Intervene: Detecting and Examining Facilitator Tendencies in Online Discussions

Anonymous ACL submission

## Abstract

The question of how and when to intervene in online discussions whether through expert or natural facilitation is a long-standing concern in the social sciences. This debate has gained new urgency with the advent of Large Language Models, which ostensibly make automated, large-scale intervention increasingly feasible. In this study, we examine human and Large Language Model tendencies of intervention, with a specific focus on positive and negative reinforcement. We then use transformers to [DETAILS TO BE ADDED]. Our results show [DETAILS TO BE ADDED].

## 1 Introduction

Deciding when to intervene in a discussion is a matter of great importance. Intervening too early or too frequently may lead to users expressing frustration at being constantly interrupted, while intervening too late may lead to a significant escalation and derailment of the discussion (Korre et al., 2025; Gao et al., 2025). However, this task remains underexplored in literature, with current studies using ground-truth labels on which users are facilitators (Chen et al., 2025; Schroeder et al., 2024; Gao et al., 2025). The task proves harder when considering that a facilitative comment is not necessarily made by dedicated facilitators; since in online environments, facilitative feedback may be primarily provided by the users themselves (Falk et al., 2024).

In this study we focus on the following Research Question (RQ):

**RQ.** *Can we reliably predict when an intervention should occur?*

We investigate different ways of defining and detecting facilitative interventions from prior Social Science and Natural Language Processing (NLP) literature in §2.1. We compare current approaches based on their assumptions, ease-of-use

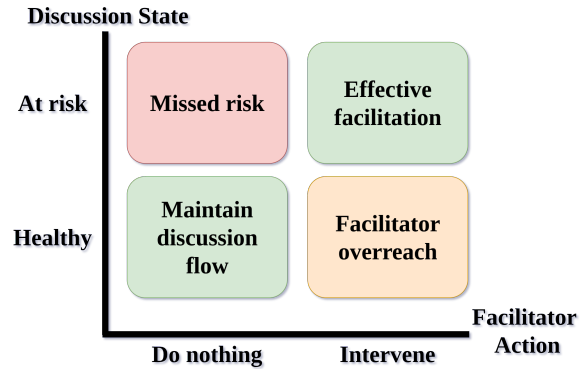


Figure 1: At any point in a discussion, the facilitator must decide whether to intervene or not. Not intervening when the situation calls for it may lead to the discussion worsening (e.g., topic derailment, escalation, missed participation). On the other hand, intervening when the discussion is already going well may lead to irritation among participants.

in research, inherent limitations, and applicability in real-world settings, finding two viable definitions. (1) A facilitative intervention should occur when a professional facilitator would have intervened (Definition 2), or (2) when the average human judges that an intervention is necessary (Definition 3).

Given the two definitions of a facilitative intervention, this RQ can be partitioned to two sub-RQs:

**RQ (a).** *Can we predict when a facilitator would have intervened?*

**RQ (b).** *Can human (or LLM) annotators agree when an intervention is necessary?*

Finally, given that current literature is split between facilitation in online, asynchronous, written exchanges and real-time spoken discussions (§??), we investigate:

**RQ (c).** *How do facilitative intervention patterns differ in written/asynchronous contexts from spoken/synchronous discussions?*

Our investigation finds that...

## 2 Related Work

### 2.1 What are facilitative interventions?

Facilitation (or conversational moderation) refers to the active participation of people in order to improve discussions (Korre et al., 2025). Relevant work assumes that facilitation is conducted by specially trained staff (Schroeder et al., 2024; Park et al., 2012), users appointed by the community (Schaffner et al., 2024), or specialized professionals (Chen et al., 2025). We identify three main definitions for what a facilitative intervention is:

**Definition 1** (Escalation). *A facilitator needs to intervene at the precise point in the discussion at which, without intervention, the discussion would derail or escalate.*

Escalation is usually measured as the point where the participants request arbitration (e.g., escalate the matter to a site moderator in Wikipedia (De Kock and Vlachos, 2021; Chang and Danescu-Niculescu-Mizil, 2019)). This definition is quite appealing for NLP researchers as the labels are easy to acquire since a ground truth exists (i.e., “escalation” can be defined in terms of moderator and user actions specific to each platform), and (2) the existence of a single, largely unambiguous ground-truth is ideal for the training and evaluation of any Machine Learning (ML)-based classifier. However, it is also inherently restrictive, as a facilitator has multiple duties that do not directly relate to preventing personal attacks, or rhetorical escalation (Korre et al., 2025). Furthermore, the datasets require that the platforms contain an unambiguous signal that a discussion has escalated, which is not often the case (Table 1). Thus, this definition is not considered in this study, but is included in our dataset (§3).

**Definition 2** (Professional facilitation). *An intervention should occur at the point where a professional facilitator would have intervened.*

Recent social science work insists that facilitators help discussions in many ways other than escalation prevention; such as summarizing discussions (Small et al., 2023), managing participation (Chen et al., 2025; Schroeder et al., 2024), and providing important information for new members (Park et al., 2012). It is generally accepted that all comments made by a facilitator should be considered facilitative interventions; for instance, all such comments are assumed to follow some social strategy (Chen et al., 2025; Schroeder et al.,

2024). These authors thus implicitly or explicitly use the definition above to distinguish between facilitative and user comments in their work.

Defining facilitative comments this way is simple and intuitive, but can only be used in domains where (1) professional facilitators exist and (2) they can be identified. These conditions exclude the vast majority of online discussions, as very few web-crawled datasets include such information. This can be seen clearly in Table 1, where the extremely large WikiConv dataset (Hua et al., 2018) can not be used for analyzing facilitation using this definition, since there is no explicit information on which user is a facilitator (in this case, a Wikipedia moderator).

**Definition 3** (Human judgment). *Facilitation should only occur when most people (annotators) agree that it is necessary.*

While some may argue that professional facilitators have more accurate judgments due to expertise, in reality most facilitation is handled by either volunteers elevated to the position by their community (Schaffner et al., 2024) or by the users themselves in an ad-hoc fashion (Falk et al., 2024). Given that humans generally hold extensive social experience because of sustained social interactions from birth, they innately understand concepts such as when to speak in a discussion (Nonomura and Mori, 2025). Thus, they are a good baseline for machine-enabled facilitation. In this case, this task is no longer about detecting or predicting whether *a facilitator would speak*, but rather the much more difficult and subjective task of whether the comment is *facilitative in nature*.

### 2.2 Spoken vs online facilitation

There currently exists a rift in facilitation research; while some previous work concerns itself with forums and online discussions (Park et al., 2012), others are exclusively focused on live, real-life, oral discussions. (Schroeder et al., 2024; Chen et al., 2025). The presence of these two sub-modalities necessitates separate experiments for either of them, since they represent different kinds of data. Posts and comments for instance are self-contained, while transcripts regularly feature interruptions, participants talking in parallel (see App. ??), and linguistic features that are necessary for coordination (such as the dash character used for signaling an interruption). They may also feature fundamentally different kinds of facilita-

tion interventions, as hinted by stark differences in proposed facilitation schemas between online forums (Park et al., 2012), and real-life discussions (Schroeder et al., 2024; Chen et al., 2025). It is worth noting that in our case, all written discussions are asynchronous (the participants may join and leave at any time), while all spoken discussions are synchronous (the participants are all active at the same time).

## 2.3 Facilitation detection

**Dimitris. What most platforms do is to use flagging and reports to decide when to intervene. Mention Grok as a facilitative agent that works by invocation only. Briefly talk about advantages and disadvantages of current approach.**

While some recent work (Schroeder et al., 2024; Chen et al., 2025) uses LLM-as-a-judge models to annotate data in the field of facilitation, the use of such models may not be an obvious choice for the task of facilitation prediction. It has been demonstrated that LLM facilitators can not decide when they should intervene irrespective of instructions and model family/size—instead, since these models are trained as chatbot assistants, they constantly attempt to speak (Tsirmpas et al., 2026). On the other hand, the use of traditional ML and Deep Learning (DL) classifiers is often used in the context of escalation prediction (Chang and Danescu-Niculescu-Mizil, 2019; Yuan and Singh, 2023).

## 3 Dataset

### 3.1 Description

Korre et al. (2025) provide a list of datasets relating to discussion facilitation which we combine to create a multi-domain dataset focused around facilitation, called the “Prosocial and Effective Facilitation in Konversations (PEFK)” dataset (Table 1). PEFK is the first dataset to compile information on facilitative interventions from both the domain of online forums, and spoken (debates, radio, and deliberative) discussions. Furthermore, our dataset standardizes discussion content, metadata, metrics, and labeling schemes according to a simple, unified schema, making it suitable as a benchmark on facilitator interventions.<sup>1</sup> See

<sup>1</sup>Specifically, interventions following the assumption of professional facilitators (Definition 2). Due to the extremely large size of the dataset, following human judgment to mark facilitative comments (Definition 3) would be prohibitively expensive.

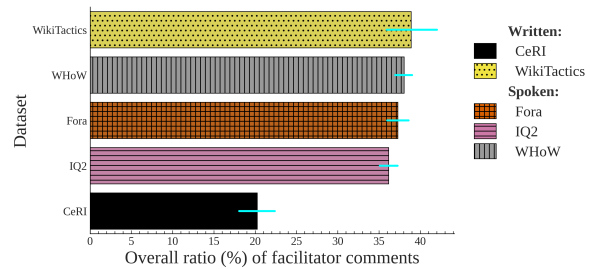


Figure 2: Percentage of moderated comments for each of the sub-datasets of PEFK. Bars denote 95% confidence intervals. Despite the different kinds of facilitators in online platforms and various real-life-discussion domains, the overall rate of facilitation remains similar across datasets.

App. B for more information on preprocessing and the schema used.

Figure 2 shows the percentage of comments made by facilitators for each of our datasets. In almost all datasets, a third of the comments are made by facilitators.<sup>2</sup>

### 3.2 Spoken vs. written datasets

Figures 3 show some of the structural differences of spoken and written datasets. Spoken datasets feature a significantly larger number of comments per discussion, since instances where the participants talk over each other, or interrupt the other are transcribed as separate comments. Additionally, spoken datasets feature a large amount of comments with very few words, since a lot of utterances are short but necessary for social coordination (e.g., “Thank you”, “Yes”, “I agree”).

## 4 Predicting facilitator comments

## 5 Seeking facilitative comments

### 5.1 Why use humans?

**Merits.** Recent research states that a fully-automated facilitation is still out of our reach as LLMs tend to be very intruding and repetitive, and fixate on the moderation side of facilitation (Tsirmpas et al., 2026; Korre et al., 2025). Human facilitators, on the other hand, can step in the conversation with a more empathetic outlook, take into deep consideration the context of the conversation, while also being more sensitive to the differences and causes of certain social behaviors.

<sup>2</sup>We do not include the UMOD dataset, since its percentage is based on our own thresholding (see B.1).

| Name                                    | #Comments  | #Discussions | Domain       | Format | Fac. |
|-----------------------------------------|------------|--------------|--------------|--------|------|
| WikiDisputes (2021)                     | 96,320     | 8,727        | Forum        | Text   | ×    |
| WikiTactics (2022)                      | 3,850      | 213          | Forum        | Text   | ✓    |
| WikiConv (2018)                         | 17,806,373 | 4,486,983    | Forum        | Text   | ×    |
| Conversations Gone Awry / CMV II (2019) | 40,607     | 6,691        | Forum        | Text   | ×    |
| CeRI (2012)                             | 3,700      | 132          | Forum        | Text   | ✓    |
| User Moderation (UMOD) (2024)           | 2,000      | 1,000        | Forum        | Text   | ✓    |
| Intelligence Squared 2 (IQ2) (2016)     | 34,245     | 108          | Debate       | Oral   | ✓    |
| Why How Who (WHoW) (2025)               | 25,542     | 178          | Radio / TV   | Oral   | ✓    |
| Fora (2024)                             | 39,438     | 262          | Deliberation | Oral   | ✓    |

Table 1: The datasets used in PEFK with accompanying post-processing statistics. We include a large variety of domains, and two different sources; text-only datasets, and transcripts from oral discussions. All datasets include multi-participant discussions, besides UMOD, which contains comment pairs. The **Fac.** column denotes whether information on which user is a facilitator is provided.

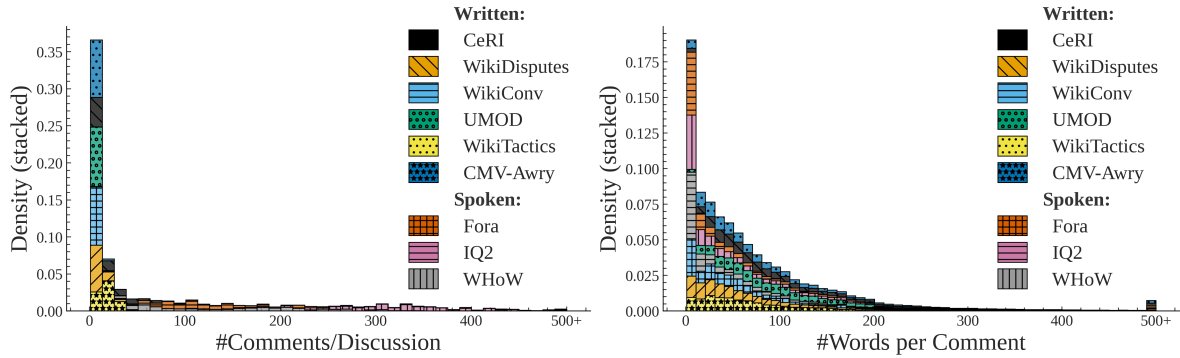


Figure 3: **Left:** Comments per discussion on the PEFK dataset. Note the significant increase in comments made in transcribed datasets such as Fora and IQ2. This deviation is explained by the fact that a single human utterance is regularly broken up to many comments in order to simulate interruptions or participants talking over each other. **Right:** Number of words per comment in all discussions on the PEFK dataset, grouped by original dataset. Comments across written datasets follow the same distribution, while spoken datasets feature a significant number of comments with very few words.

Table 2: Classifier performance trained on all datasets

| Dataset     | Precision | Recall | F1    | Support |
|-------------|-----------|--------|-------|---------|
| ceri        | 0.216     | 0.103  | 0.139 | 156     |
| fora        | 0.384     | 0.695  | 0.495 | 2692    |
| iq2         | 0.421     | 0.680  | 0.520 | 2462    |
| whow        | 0.407     | 0.637  | 0.497 | 1875    |
| wikitactics | 0.542     | 0.424  | 0.476 | 243     |

**Disadvantages.** However, recruiting human facilitators, which also indicates a sort of expertise, is not a scalable process. In terms of LLM and human annotation alignment, (Movva et al., 2024) focusing on online safety contexts, show that LLMs are quite monodimensional and larger datasets are needed to understand if LLMs can adequately represent different demographic groups. This can also be translated in the context of facilitation, when LLMs *get triggered* only when toxicity appears in the text, disregarding the encouragement

of prosocial behavior.

## 5.2 Experimental setup

Our experimental setup is guided by two key questions: Under what circumstances do facilitators intervene? and How do they intervene? For the second question, rather than attempting to provide an exhaustive list of all possible methods, we focus specifically on positive and negative reinforcement. Positive and negative reinforcement here should be understood as insights that emerged from our discussions with annotators, rather than as predefined moderation tools. In particular, positive reinforcement has often been overlooked, even though our goal in annotation and evaluation was not limited to moderation alone. Instead, we aimed to capture a broader range of interaction dynamics, including how constructive or supportive responses can be recognized and encouraged, not just how harmful content is flagged or constrained. At the same time, our analysis operates



at a higher level focused on facilitation rather than fine-grained categorization so we do not center on highly specific reinforcement types, but instead draw on initial intuitions such as their relative frequency and overall role in shaping interactions

**Data Segmentation** We split multi-party discussions from the PEFK dataset into small, defined chunks based on [discourse boundaries like topic shifts and adjacency relations]. Annotators would read only one chunk at a time rather than full conversations. This design limited access to global context while preserving local coherence within each unit. We did this to align annotation with realistic, moment-by-moment interpretation in natural dialogue. It reduces hindsight bias from later parts of the conversation influencing earlier judgments. It also prevents annotators from reconstructing overly coherent narratives that were not locally available.

**Annotator Profile.** In designing our data annotation framework, we intentionally prioritized the recruitment of expert annotators over expert moderators. This distinction is pivotal to our research design: whereas expert moderators typically engage in post hoc intervention addressing content only after its publication expert annotators facilitate a more proactive analysis of the discourse as it unfolds. However, acknowledging the importance of moderation experience we also include a professional moderator into the workflow to provide a more just comparison.

We operate under the premise that the majority of discussion facilitation and moderation is conducted organically by the participants themselves through user-driven flagging mechanisms or actual dialogue.

We recruited 10 expert annotators for the task of reinforcement detection and classification. All annotators had extensive experience with NLP annotation tasks. The gender distribution was relatively balanced with 4 males and 6 females, while their ages had a median of 25 years. Before the commencement of the annotation process, the annotators completed an agreement form in compliance with GDPR ethical requirements. They were compensated in accordance with Greek legislation.

**Annotation Procedure.** As described in the previous sections of this article, the goal is to understand *when intervention is needed* as well as *what*

| Category         | Humans | Models |
|------------------|--------|--------|
| Positive         | 0.50   | 0.50   |
| Negative         | 0.58   | 0.56   |
| No-Reinforcement | 0.47   | 0.46   |

Table 3: IAA (Fleiss  $\kappa$ ) comparison within the human annotator group, model group, and both humans and models.

*kind of intervention is needed.* We select a total of 1212 random discussions from all datasets (with a uniform distribution) and select a random section with length 6 comments or less from each. Due to the disagreement with regard to a taxonomy of types of interventions (we should refer the reader to a specific point in the text), we opted for examining only on one type of intervention, that is reinforcement. Since reinforcement [annotation] can be a subjective task (point out to a reference), we adopt an ordinal scale approach, where the annotators rate each dialogue for either type of reinforcement (positive or negative), as well as the possibility of no reinforcement using a scale from 1 to 5. The full annotation guidelines can be found in the Appendix [to add the ref].

### 5.3 Results

Apart from some *negligible* exceptions, human annotators very cautious when it comes to intervening, with most often selecting to not intervene (no reinforcement). When it comes to whether they choose positive or negative reinforcement, the discrepancy is not that great, with positive reinforcement being selected slightly more often. However, it must be noted (we need to connect this with the previous results), that despite the slightly higher frequency of positive reinforcement, they agree more on where intervene with negative reinforcement. We also run a small scale IAA analysis looking at the oral datasets (debates, and ..) in comparison to the written datasets (forums) and the difference in IAA is minimal with 30% and 33% Fleiss  $\kappa$  respectively.

### 5.4 Could we use LLMs instead of annotators?

The models on the other hand, are more consistent, with overall higher intervention frequency than you humans (we need to connect this with the fact that models very often become too intrusive). Also, they are more liberal with the intervention types, with the most frequent one be-



Figure 4: Pairwise Cohen's  $\kappa$  per class.

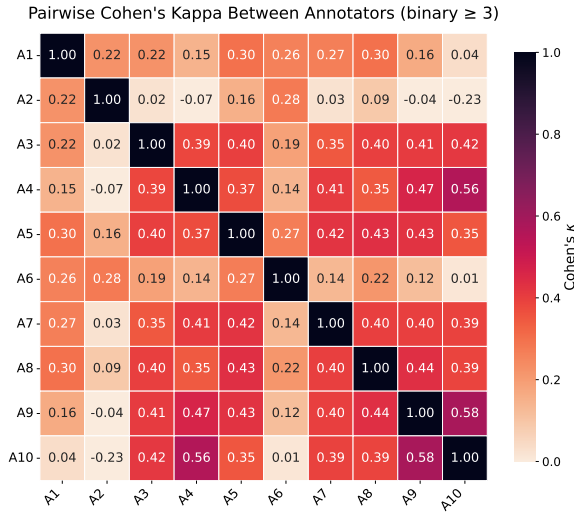


Figure 5: Heatmap of pairwise Cohen's  $\kappa$  scores.

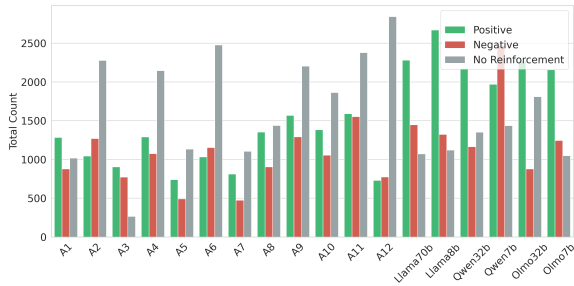


Figure 6: Type of reinforcement the human and LLM annotators choose to intervene with.

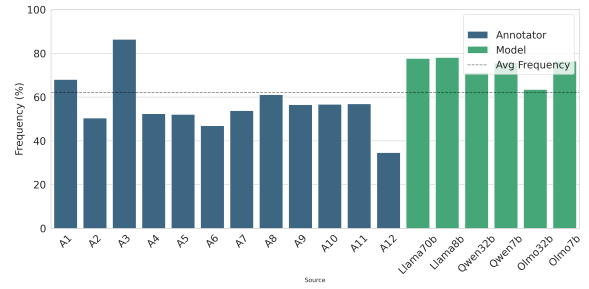


Figure 7: Frequencies of human and LLM annotator interventions.

ing with the discussion, or explicitly having a facilitator role. We initially hypothesized that this setup would lead to the models being less prone to intervening, since they are no longer instructed to make the discussion better as a facilitator.<sup>3</sup> However, the rates of interventions seem to be the same as the ones reported by the authors, suggesting that the *over-eagerness to intervene is unlikely to be solvable by simply using prompting*. Therefore, we believe that out-of-the-box LLMs can not be used for facilitation.

## 6 Conclusion

Humans are good but expensive. Our research is good. We wasted a good amount of money on this, please accept this.

## 7 Limitations

None, we are awesome.

<sup>3</sup>The fact that LLMs are biased to always respond was hypothesized by the original authors to be the main reason this phenomenon occurs.

## 8 Ethical Considerations

All experiments and data have passed through official ethical approval by an Ethics Committee, and are compatible with GDPR (European Parliament and Council of the European Union) and the EU AI Act (European Parliament and Council of the European Union, 2024). The annotators were compensated fairly with a monthly rate of 1000 and signed informed consent forms before beginning annotation and were allowed to skip uncomfortable tasks at any point, with no required justification.

In order to help research in the field of facilitation, we openly release our dataset under a CC-BY-SA license, and the code that compiles it under GPLv3.

## References

- Jonathan P. Chang and Cristian Danescu-Niculescu-Mizil. 2019. [Trouble on the horizon: Forecasting the derailment of online conversations as they develop](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 4743–4754, Hong Kong, China. Association for Computational Linguistics.
- Ming-Bin Chen, Lea Frermann, and Jey Han Lau. 2025. [WHoW: A cross-domain approach for analysing conversation moderation](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2091–2126, Albuquerque, New Mexico. Association for Computational Linguistics.
- Christine De Kock, Tom Stafford, and Andreas Vlachos. 2022. [How to disagree well: Investigating the dispute tactics used on wikipedia](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 3824–3837, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Christine De Kock and Andreas Vlachos. 2021. [I beg to differ: A study of constructive disagreement in online conversations](#). In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 2017–2027, Online. Association for Computational Linguistics.
- European Parliament and Council of the European Union. [Regulation \(EU\) 2016/679 of the European Parliament and of the Council](#).
- European Parliament and Council of the European Union. 2024. [Regulation \(eu\) 2024/1689](#)

of the european parliament and of the council of 13 june 2024 laying down harmonised rules on artificial intelligence (artificial intelligence act). <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>. Official Journal of the European Union, L 2024/1689.

- Neele Falk, Eva Vecchi, Iman Jundi, and Gabriella Lapesa. 2024. [Moderation in the wild: Investigating user-driven moderation in online discussions](#). In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 992–1013, St. Julian’s, Malta. Association for Computational Linguistics.
- Rena Gao, Ming-Bin Chen, Lea Frermann, and Jey Han Lau. 2025. [Moderation matters: Measuring conversational moderation impact in English as a second language group discussion](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 2070–2095, Vienna, Austria. Association for Computational Linguistics.
- Yiqing Hua, Cristian Danescu-Niculescu-Mizil, Dario Taraborelli, Nithum Thain, Jeffery Sorensen, and Lucas Dixon. 2018. [Wikiconv: A corpus of the complete conversational history of a large online collaborative community](#). In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2818–2823, Brussels, Belgium. Association for Computational Linguistics.
- Katerina Korre, Dimitris Tsirmpas, Nikos Gkoumas, Emma Cabalé, Danai Myrtzani, Theodoros Evgeniou, Ion Androutsopoulos, and John Pavlopoulos. 2025. [Evaluation and facilitation of online discussions in the LLM era: A survey](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 24443–24462, Suzhou, China. Association for Computational Linguistics.
- Rajiv Movva, Pang Wei Koh, and Emma Pierson. 2024. [Annotation alignment: Comparing LLM and human annotations of conversational safety](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 9048–9062, Miami, Florida, USA. Association for Computational Linguistics.
- Ryota Nonomura and Hiroki Mori. 2025. [Who speaks next? multi-party ai discussion leveraging the systematics of turn-taking in murder mystery games](#). *Frontiers in Artificial Intelligence*, Volume 8 - 2025.
- Joonsuk Park, Sally Klingel, Claire Cardie, Mary Newhart, Cynthia Farina, and Joan-Josep Vallbé. 2012. [Facilitative moderation for online participation in erulemaking](#). In *Proceedings of the 13th Annual International Conference on Digital Government Research*, dg.o ’12, page 173182, New York, NY, USA. Association for Computing Machinery.
- Perspective API. 2026. [Perspective api](#). Accessed: 2026-02-24.

Brennan Schaffner, Arjun Nitin Bhagoji, Siyuan Cheng, Jacqueline Mei, Jay L Shen, Grace Wang, Marshini Chetty, Nick Feamster, Genevieve Lakier, and Chenhao Tan. 2024. "community guidelines make this the best party on the internet": An in-depth study of online platforms' content moderation policies. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, CHI '24, New York, NY, USA. Association for Computing Machinery.

Hope Schroeder, Deb Roy, and Jad Kabbara. 2024. *Fora: A corpus and framework for the study of facilitated dialogue*. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13985–14001, Bangkok, Thailand. Association for Computational Linguistics.

Christopher T. Small, Ivan Vendrov, Esin Durmus, Hadjar Homaei, Elizabeth Barry, Julien Cornebise, Ted Suzman, Deep Ganguli, and Colin Megill. 2023. *Opportunities and risks of llms for scalable deliberation with polis*. *ArXiv*, abs/2306.11932.

Dimitris Tsirmpas, Ion Androutsopoulos, and John Pavlopoulos. 2026. *Designing synthetic discussion generation systems: A case study for online facilitation*. *Preprint*, arXiv:2503.16505.

Jiaqing Yuan and Munindar P. Singh. 2023. *Conversation modeling to predict derailment*. *Proceedings of the International AAAI Conference on Web and Social Media*, 17(1):926–935.

Justine Zhang, Ravi Kumar, Sujith Ravi, and Cristian Danescu-Niculescu-Mizil. 2016. *Conversational flow in Oxford-style debates*. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 136–141, San Diego, California. Association for Computational Linguistics.

## A Acronyms

**AI** Artificial Intelligence

**NLP** Natural Language Processing

**ML** Machine Learning

**RQ** Research Question

**PEFK** Prosocial and Effective Facilitation in Konversations

**DL** Deep Learning

## B Dataset details

### B.1 Preprocessing

#### General

- We exclude comments with no text.
- We exclude discussions with less than two distinct participants.
- We exclude discussions which are common between wiktactics and wikiconv as well as wikidisputes and wikiconv.
  - There may be duplicate discussions between wikidisputes and wiktactics, but we allow them since they feature complementary information.

#### Wikiconv

- The Wikiconv corpus does not contain information about which user is a moderator/facilitator. Therefore, all comments relating to Wikiconv are tagged as non-moderators.
- Additionally, we follow the instructions of the original researchers (Hua et al., 2018), and select only discussions which have at least two comments by different users.
  - Wikipedia (thankfully) does not track users who log in with only an IP address (in the original dataset, their user\_id is always set to 0 and their username is of the form 211.111.111.xxx. We consider each such username to be a separate user.
  - Due to the size of the dataset, we have to partially load it during preprocessing. Thus, there is a small chance every 100,000 records that a discussion is marked as a false negative and a part of it gets discarded.
  - We only include English comments in the final dataset. We use a small, efficient library (py3langid) for language recognition, due to the large size of Wikiconv. Non-English comments are discarded before selecting valid discussions (see point above).

#### Wikitactics

- We infer facilitative actions by whether the comment belongs in any of the following categories:
  - Asking questions
  - Coordinating edits
  - Providing clarification
  - Suggesting a compromise
  - Contextualisation
- The above tactics are a subset of the Coordinative labels used in the WikiTactics paper.



They were selected because they are not used necessarily on 1-1 discussions; they could reasonably be applied by third-party participants. Contrast them with other Coordinative labels such as “Conceding/recanting” and “I don’t know”.

## Wikidisputes

- Since only 0.03% of the comments in the dataset are made by moderators, we mark the dataset as not supporting moderation.

## UMOD

- Facilitative actions are marked as a gradient from 0 (no facilitation) to 1 (full facilitation). We adopt a threshold of 0.75 to consider an action as facilitative, with more than 50% annotator agreement (measured as entropy in the original dataset).

## CMV-AWRY2

- We mark a discussion as escalated when the derailment value (from the official dataset) is in the 60th upper percentile.
- We remove deleted (“[deleted]”) comments.

## B.2 Schema

Table 4 details the information available for each comment in the dataset. Some information, such as whether the person talking is a moderator / facilitator, is available only on some sub-datasets. Also note that the `is_moderator` column may hold different definitions of what we consider as moderator / facilitator. UMOD for instance considers facilitative comments made by users, while others such as Fora consider a comment facilitative if it was made by a dedicated facilitator.

## C Expert facilitators vs. expert annotators

A11 does not exhibit the characteristics one would expect from a distinct expert annotator, but instead behaves as a highly balanced and representative member of the group. In contrast to other annotators who show clear directional biases either favoring positive reinforcement (e.g., A1, A7, A8) or leaning negative (e.g., A2, A6) A11 distributes judgments almost evenly between positive (28.80%) and negative (28.12%) labels, resulting in a near-zero difference (0.69 percentage points). At the same time, A11’s intervention rate (56.92%) is almost identical to the group average, indicating

that it is neither more strict nor more lenient in deciding when to intervene. When considered alongside the moderate correlation scores with other annotators (mean 0.38), this pattern suggests that A11 does not consistently align more strongly with the group than others do, nor does it systematically deviate in a principled way. Instead, A11 appears to occupy the center of the annotator space: its behavior closely reflects the aggregate tendencies of the group without amplifying any particular bias. Consequently, the data do not support interpreting A11 as a clear expert or gold-standard annotator; rather, A11 is better characterized as a calibrated, balanced annotator whose decisions approximate the overall distribution of judgments in the dataset.

## D Prompts used

### D.1 Yes/no intervention

You are an expert annotator specializing in online moderation and facilitation. You will be presented with a comment and the comments made before it. Choose whether you decide to intervene at this point in the discussion.

Key Rules for Annotating:

**Look at Intent and Impact** Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

**Use the Context** Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context and vice versa.

**Judge Behavior, Not Opinions** The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.

**Remain Consistent** Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.

**Output** Respond whether you would intervene in this discussion as a facilitator on a scale of 1-5, where 1 is “I’m sure I don’t need to intervene”, and 5 is “I’m sure I need to intervene”

### D.2 Positive vs. negative reinforcement

You are an expert facilitator. You will be presented with a discussion. Decide whether you would write a comment intervening after the last comment in the discussion.

**The Core Rating Task**

| Name                 | Type   | Description                                                                                                                                              |
|----------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| conv_id              | string | The discussion’s ID. Comments under the same discussion refer to the same discussion ID.                                                                 |
| message_id           | string | The message’s (comment’s) unique ID.                                                                                                                     |
| reply_to             | string | The ID of the comment which the current comment responds to. nan if the comment does not respond to another comment (e.g., it’s the Original Post (OP)). |
| user                 | string | Username or hash of the user that posted the comment                                                                                                     |
| is_moderator         | bool   | Whether the user is a facilitator.                                                                                                                       |
| moderation_supported | bool   | True if the moderation labels are directly computed from the original dataset                                                                            |
| escalated            | bool   | A discussion-level measure denoting discussions which have been derailed                                                                                 |
| escalation_supported | bool   | True if the escalation labels are directly computed from the original dataset                                                                            |
| text                 | string | The contents of the comment                                                                                                                              |
| dataset              | string | The dataset from which this comment originated from                                                                                                      |
| notes                | JSON   | A dictionary holding notable dataset-specific information                                                                                                |
| toxicity             | float  | The “toxicity” score given to the comment by the <a href="#">Perspective API</a>                                                                         |
| severe_toxicity      | float  | The “severe toxicity” score given to the comment by the <a href="#">Perspective API</a>                                                                  |

Table 4: The full set of columns used for each comment in the [PEFK](#) dataset.

| Annotator   | Correlation with A11 |
|-------------|----------------------|
| A6          | 0.492                |
| A10         | 0.464                |
| A2          | 0.432                |
| A8          | 0.430                |
| A1          | 0.396                |
| A9          | 0.385                |
| A4          | 0.383                |
| A3          | 0.363                |
| A12         | 0.301                |
| A7          | 0.279                |
| A5          | 0.249                |
| <b>Mean</b> | <b>0.380</b>         |
| <b>Min</b>  | <b>0.249</b>         |
| <b>Max</b>  | <b>0.492</b>         |

Table 5: Correlation of Expert Moderators with Other Annotators

For every comment, you must write out a response of 1 (uncertain) to 5 (certain):

#### Positive Reinforcement

Defined as an intervention whose purpose is to encourage a specific behavior. For example, a moderator may ask people to participate, ask a person to elaborate etc.

Positive reinforcement is not related to positive speech; the moderator may be strict or even rude when attempting to encourage specific behaviors

(e.g., “I will not tolerate any further discussion on this topic. It’s been decided and we’re moving on.”).

Rate this when a conversation demonstrates behavior that should be encouraged or reinforced in a synchronous exchange. Your task is not simply to identify positive behaviors, but to identify moments where reinforcing them would help sustain or build constructive conversational momentum.

This includes, but is not limited to:

- Helpful or clarifying explanations
- Respectful disagreement
- Constructive feedback
- Polite, empathetic, or supportive tone
- Efforts to de-escalate conflict
- Any contribution that meaningfully improves the quality or direction of the conversation

#### Negative Reinforcement

The purpose of negative reinforcement is to discourage unwanted or disruptive behaviors. This is correlated, but not necessarily closely related to threats or enforcement of disciplinary action.

Negative reinforcement is not related to negative speech; the moderator may be kind and diplomatic when discouraging behaviors (e.g., “I’ve received multiple complaints about your behavior on

| Annotator | Negative (%) | No Reinforcement (%) | Positive (%) | Intervention Rate (%) | Pos - Neg (pp) |
|-----------|--------------|----------------------|--------------|-----------------------|----------------|
| A1        | 27.61        | 31.99                | 40.40        | 68.01                 | 12.78          |
| A2        | 27.65        | 49.62                | 22.73        | 50.38                 | -4.92          |
| A3        | 39.77        | 13.65                | 46.57        | 86.35                 | 6.80           |
| A4        | 23.82        | 47.59                | 28.59        | 52.41                 | 4.77           |
| A5        | 20.78        | 47.95                | 31.27        | 52.05                 | 10.50          |
| A6        | 24.73        | 53.11                | 22.16        | 46.89                 | -2.57          |
| A7        | 19.84        | 46.21                | 33.95        | 53.79                 | 14.11          |
| A8        | 24.47        | 38.93                | 36.60        | 61.07                 | 12.13          |
| A9        | 25.52        | 43.51                | 30.97        | 56.49                 | 5.45           |
| A10       | 24.52        | 43.32                | 32.15        | 56.68                 | 7.63           |
| A11       | 28.12        | 43.08                | 28.80        | 56.92                 | 0.69           |
| A12       | 17.81        | 65.40                | 16.79        | 34.60                 | -1.01          |

Table 6: Behavioral Summary of Annotators (Percentages)

this forum. Please refrain from making personal attacks and inflammatory comments.”).

Rate this when the comment reflects harmful, disruptive, or inappropriate behavior that should be discouraged, as well as cases where some participants dominate the conversation, or when the conversation starts to go off topic.

Other examples include:

- Hostile, dismissive, or disrespectful language
- Personal attacks or insults
- Repeated interruptions or monopolizing the exchange
- Escalating conflict rather than resolving it
- Contributions that reduce the overall quality of the conversation

#### No Reinforcement

Rate this when the comment is neutral, unremarkable, or does not affect the tone or direction of the conversation.

This includes:

- Basic factual statements
- Clarifying questions
- Minimal or vague comments
- Harmless but non-notable contributions

#### Key Rules for Annotating

##### Look at Intent and Impact

Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

##### Use the Context

Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context and vice versa.

##### Judge Behavior, Not Opinions

The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.

#### Remain Consistent

Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.

#### Output

Respond whether you would intervene in this discussion as a facilitator.

Always respond with the following format and nothing else:

Positive: 1-5

Negative: 1-5

No Reinforcement: 1-5

#### D.3 Single-comment facilitation

You are an expert annotator specializing in online moderation and facilitation. You will be presented with a comment and the comments made before it. Choose whether you decide to intervene at this point in the discussion.

Key Rules for Annotating:

**Look at Intent and Impact** Consider what the commenter seems to be trying to do, and how their words might affect others in the thread.

**Use the Context** Always check the surrounding messages. A comment that looks fine on its own might escalate tension when viewed in context and vice versa.

**Judge Behavior, Not Opinions** The label should never depend on whether you personally agree with the comment. Focus on tone, clarity, constructiveness, and how it influences the conversation.

**Remain Consistent** Apply the same standards across all comments. Avoid guessing or over-interpreting; use what is present in the text.

**Output** Respond whether you would intervene in this discussion as a facilitator on a scale of 1-

803 5, where 1 is "I'm sure I don't need to intervene",  
804 and 5 is "I'm sure I need to intervene"